

Supplementary Material S3: List of Studies Excluded at Full-Text Stage with Reasons

Studies excluded after full-text review (n = 4)

#	Study	Reason for Exclusion	Detail
1	Malik A et al. (2023) “Outcomes of Ultrasound Guided Peripheral Intravenous Catheters Placed in the Emergency Department and Factors Associated with Survival.” <i>Open Access Emergency Medicine</i> .	Single-arm study, no comparator	Descriptive study of 388 USIV placements only; no landmark/palpation comparison group. Does not meet inclusion criterion #1.
2	Avelar AF et al. (2013) “Assertiveness and dwell times with USG in children.”	Non-English companion report	Portuguese-language companion report of the same trial (NCT00930254) now included via its English 2015 report (Avelar 2015 [5]); excluded only as a non-English duplicate report, handled identically to the duplicate Kleidon/EPIC reports below.
3	Unpublished manuscript — “Effectiveness of Ultrasound-Guided Versus Traditional IV Insertion by Oncology Nurses in Adult Population Receiving Systemic Anti-cancer Therapy”	Not published	Manuscript not published at time of search.
4	Kleidon TM et al. (2023) “Ultrasound-guided PIVC insertion: a randomised controlled trial protocol.” <i>British Journal of Nursing</i> .	Protocol only, no results	Protocol paper for the EPIC trial; same trial as Kleidon 2025 (JAMA Pediatrics). No post-insertion outcome data reported.

Note: Two records initially flagged by a single reviewer — the ICU-nurses trial registration (ClinicalTrials.gov NCT03745209) and the Kleidon et al. conference abstract (Journal of Vascular Access, 2022, an overlapping report of the EPIC trial) — were adjudicated out at the title/abstract

screening stage and are therefore counted among the title/abstract exclusions (within the 1,340), not the full-text exclusions.

Full-text articles retrieved and included (previously not retrieved)

#	Study	Rayyan ID	Status
1	Refosco M et al. (2024) — USG deep veins vs blind superficial veins in children 0-18y	486160202	Retrieved and included (confounded by catheter type)
2	Cottrell JT et al. (2021) — USG increases catheter dwell time in children	486160348	Retrieved and included

Full-text articles retrieved after initial search (subsequently included)

#	Study	Rayyan ID	Status
1	Desai K et al. (2018) — Longevity and Complication Rates of Ultrasound Guided vs Traditional PIVC in a Paediatric ED. <i>JAVA</i> 23(3):149-154.	486161209	Retrieved April 2026 via interlibrary loan; included (classified as catheter-length confounded; excluded from primary pooling, retained in narrative synthesis)
2	Nishizawa T et al. (2020) — Nurse-Performed Ultrasound-Guided Technique for Difficult Peripheral IV Access in Critically Ill Patients. <i>JAVA</i> 25(2):34-39.	486161498	Retrieved April 2026 via interlibrary loan; included (5th RCT; extravasation data contributed to narrative synthesis)